

Saturated fatty acid metabolism is key link between cell division, cancer, and senescence in cellular and whole organism aging

Cellular senescence is an in vivo and in vitro phenomenon, accompanied by physiological changes including cessation of division and disturbances of organelle structure and function. Review of the literature was undertaken to determine whether there is evidence that whole organism aging and cell senescence share a common initiation pathway. In vivo aged cells of different lineages, including aged T lymphocytes, show high expression of the INK4A-p16 gene. In cell culture when telomeres are shortened past a key length or state, the Arf/Ink gene system (p16/p14 humans, p16/p19 mice) switches on and activates p53, which suppresses further cell division. The p53 gene is a key tumor suppressor and its deletion or mutation allows cancerous growth. The switching on of p53 also causes changes in fatty acid metabolism, especially down-regulation of both fatty acid synthase and stearoyl-CoA (Δ -9) desaturase. The co-suppression of these genes together with enhanced uptake of extracellular fatty acids, leads to raised levels of cellular palmitate and induction of either apoptosis or senescence. In senescent cells, the fatty acid composition of the cellular membranes alters and leads to changes in both structure and function of organelles, especially mitochondria. Animal models of accelerated aging exhibit repression of stearoyl-CoA desaturase activity while anti-aging calorie restriction stimulates the same enzyme system. It is concluded that aging in cells and whole organisms share a common initiation pathway and that cellular senescence is protective against cancer. Healthy longevity is likely to be most enhanced by factors that actively suppress excessive cell division.